MSC 7208 DATA MINING · PROJECT 2: GRAPH MINING

Graph Mining of the Facebook Large Page–Page Network

Centrality · Community Detection · Link Prediction — at scale on 22,470 nodes

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PROJECT SCOPE

Four analyses on one network

TASK 1

Construction & Exploration

Order, size, density, connectivity and the degree distribution.

TASK 2

Centrality Analysis

Four measures of influence; identify the most important pages.

TASK 3

Community Detection

Louvain & Girvan–Newman; compare partition quality by modularity.

TASK 4

Link Prediction

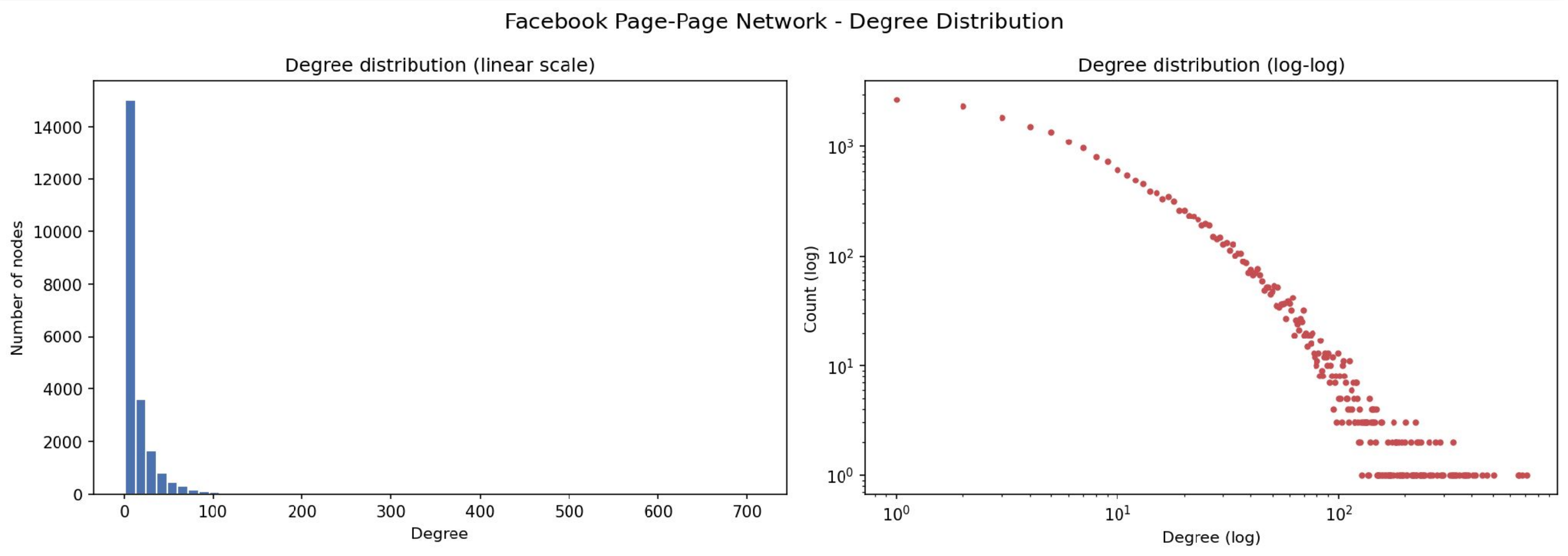
Predict missing edges from topological features; evaluate with AUC.

A single, sparse, hub-dominated network

22,470 Nodes (Facebook pages)	**170,823** Edges (mutual likes)	**1** Connected component (100%)
6.77×10^{-4} Graph density	**15.2** Mean degree	**709** Maximum degree

Why it matters: the wide degree range and the single giant component are the signature of a scale-free social graph — most pages have a handful of links while a few hubs have hundreds.

The heavy tail, made visible



Reading the plot

Linear (left):

a spike at low degree — most pages.

Log–log (right):

a near-straight line ⇒ approximately power-law, $P(k) \sim k^{-\gamma}$.

Four lenses on “influence”

Degree

Local connectivity — how many direct links.

Betweenness

Bridging — how often on others’ shortest paths.

Closeness

Reach — inverse mean distance to all nodes.

PageRank

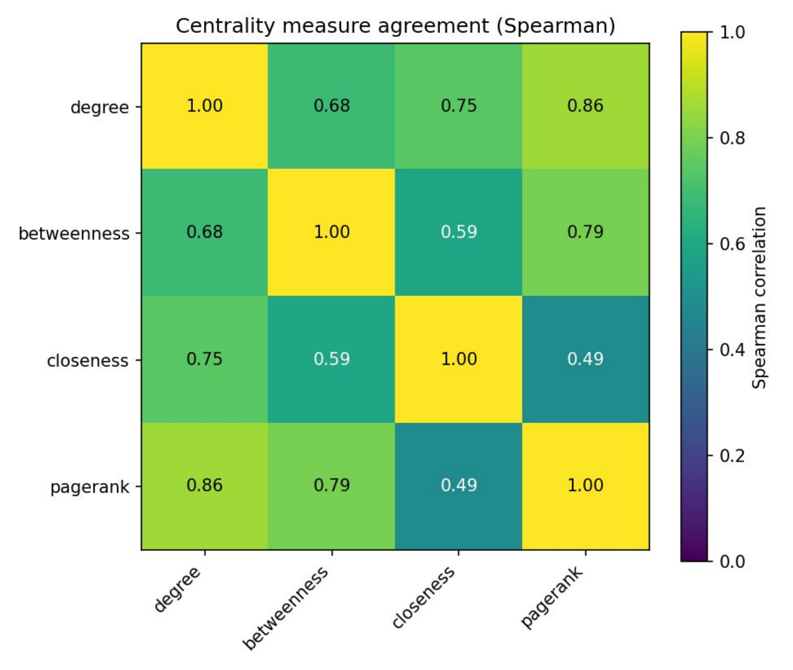
Recursive prestige — linked-to by important pages.

Scale problem: exact betweenness and closeness are O(V·E) — impractical at 22k nodes. We use pivot-sampled, parallelised estimators, validated to ≈0.99 rank-correlation with exact.

Who is most influential — and the measures disagree

Measure	#1	#2	#3
Degree	U.S. Army	The White House	Obama White House
Betweenness	Facebook	Barack Obama	The White House
Closeness	Facebook	Barack Obama	Obama White House
PageRank	Facebook	Sir Peter Bottomley	Harish Rawat

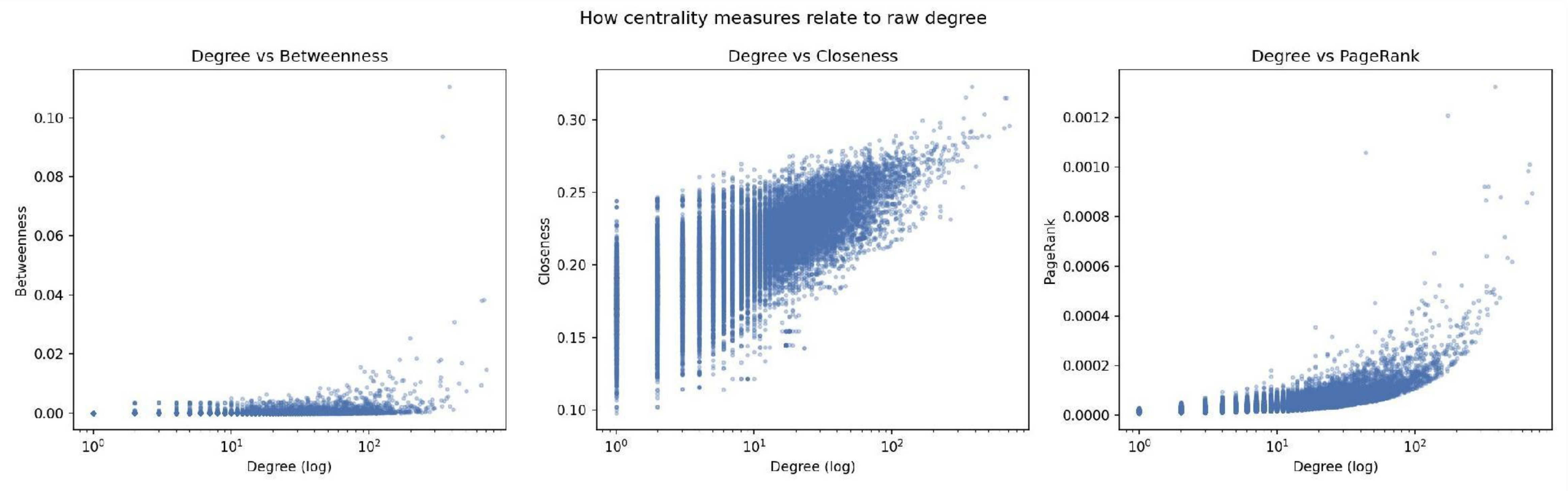
Composite (mean rank): The White House · Facebook · Obama White House



Spearman ρ — closeness & PageRank agree least (0.49)

Interpretation: Facebook bridges the whole network (top betweenness); the U.S. Army is locally hyper-connected (top degree). Different maths, different 'important'.

Each measure scales with degree differently



Betweenness

Flat then explosive; only the few mega-hubs bridge the network.

Closeness

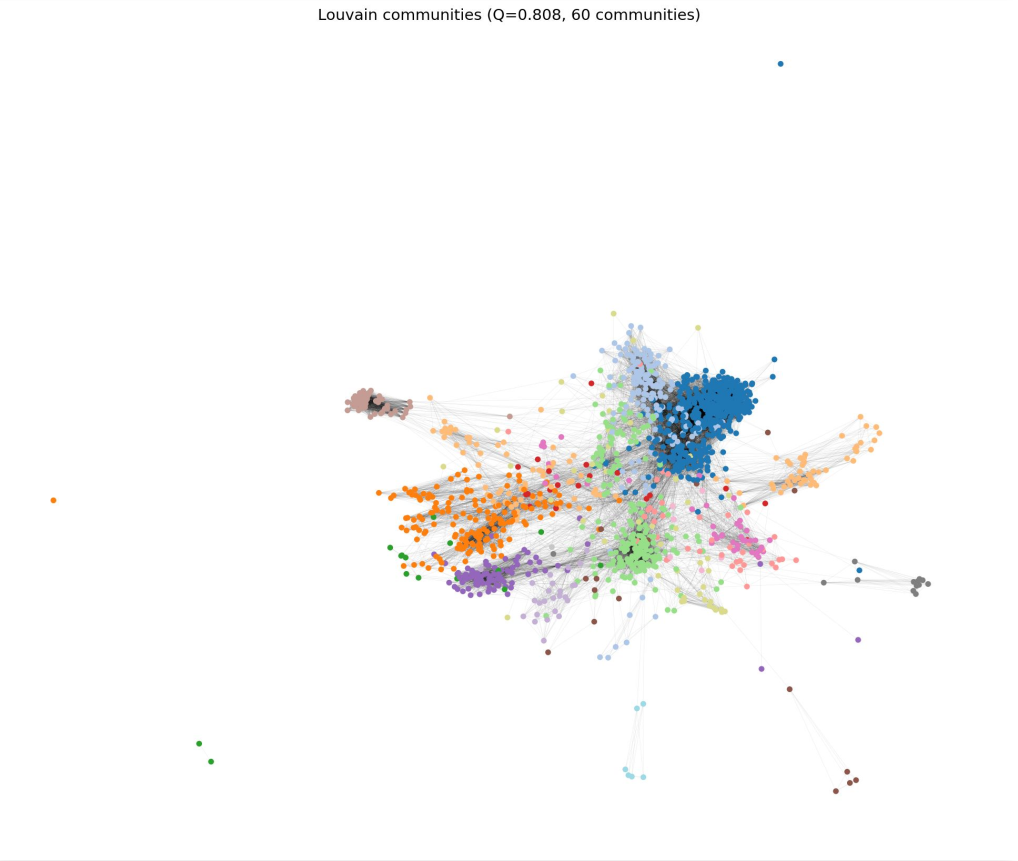
Rises steadily; more links generally means closer to everyone.

PageRank

Curves upward; importance compounds through important neighbours.

TASK 3 · COMMUNITY DETECTION

Strong community structure: Q = 0.808



Louvain communities on the full graph

60 communities found by Louvain (full graph)

0.808 modularity Q — well above the 0.7 ‘strong’ threshold

3,342 nodes in the largest community

Modularity $Q = (1/2m)\ \Sigma\ [A_{ij} - k_i k_j/2m]\ \delta(c_i,c_j)$

TASK 3 · WHICH ALGORITHM WINS

Louvain vs Girvan–Newman — a fair, same-subgraph test

Louvain

0.290

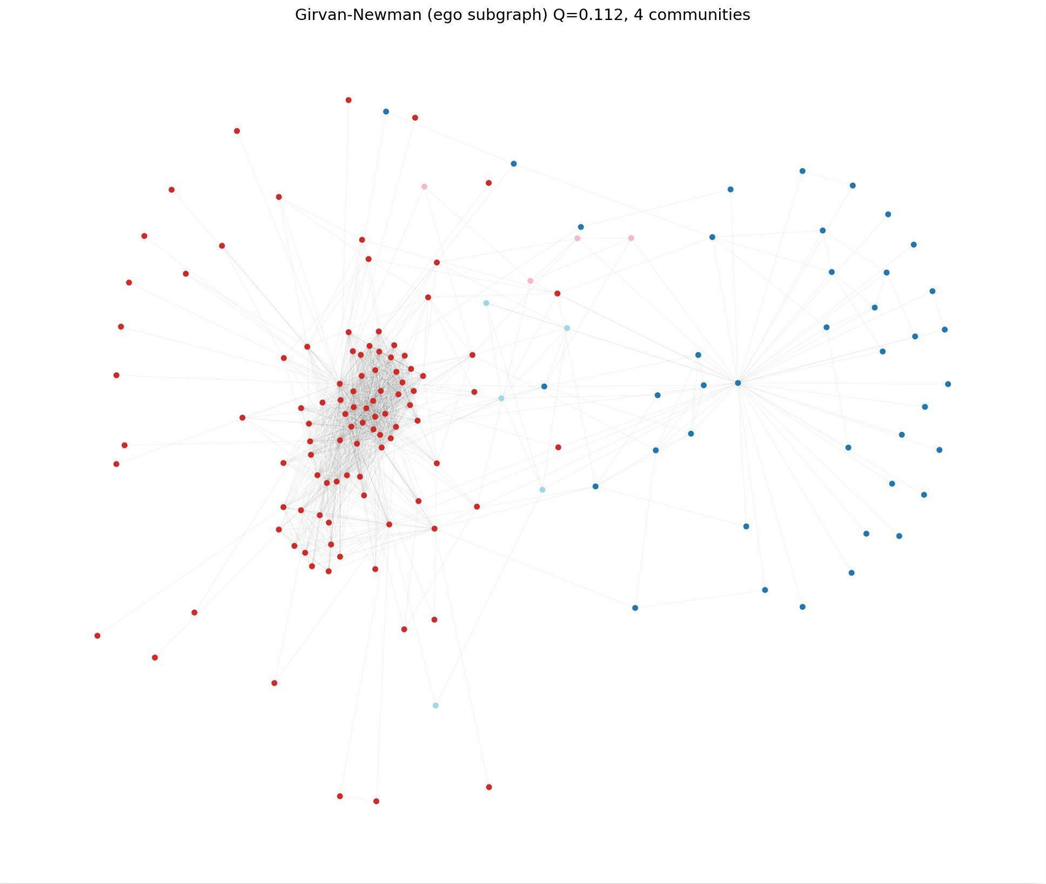
4 communities
(150-node subgraph)

Girvan–Newman

0.112

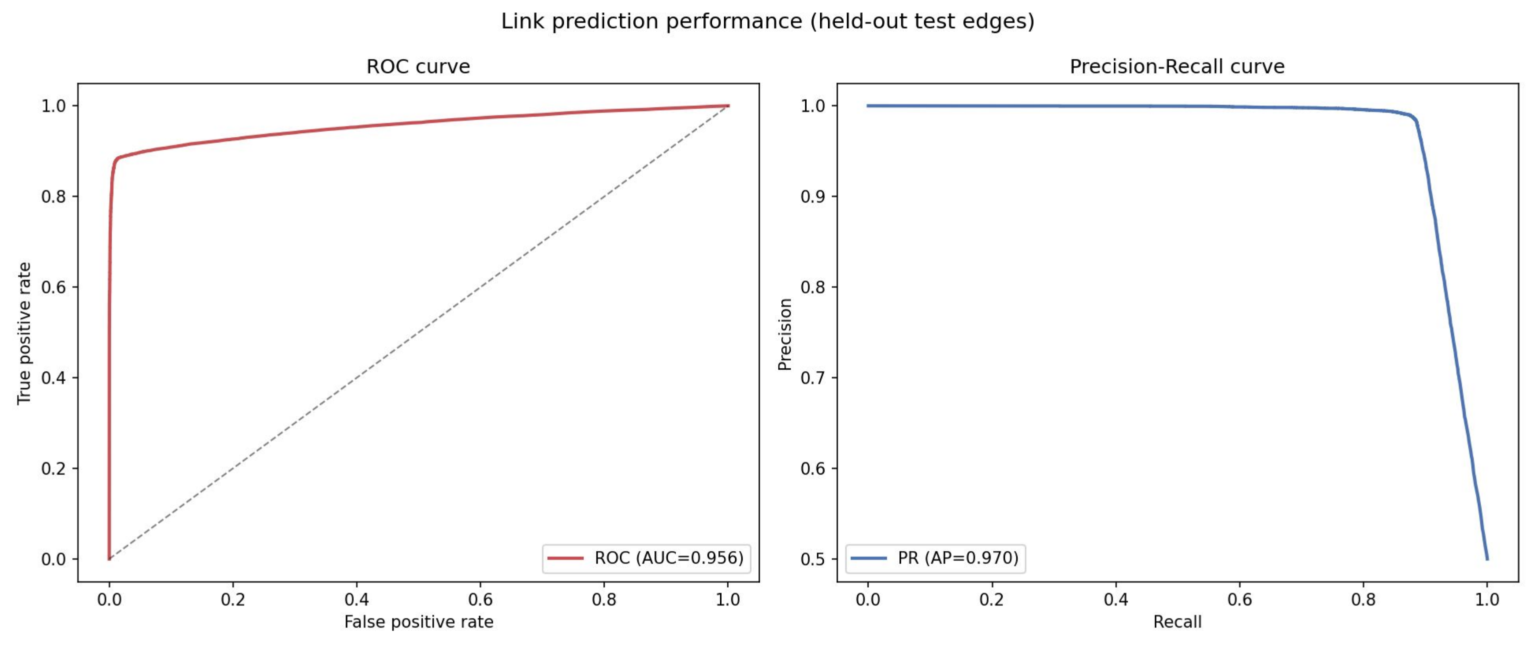
4 communities
(150-node subgraph)

Louvain wins. Double the modularity on identical input — and it scales to all 22k nodes, which GN cannot.



Girvan–Newman partition (Q=0.112): one dense red core plus scattered nodes — visibly weak structure

Missing links are highly predictable: AUC 0.956



0.956 ROC-AUC

0.970 PR-AUC (AP)

0.933 Accuracy

Logistic regression on 4 topological scores. Adamic–Adar & resource-allocation dominate — shared selective neighbours predict links far better than raw popularity.

ENGINEERING JUDGEMENT

Traps we hit — and how we countered them

The real bottleneck wasn't betweenness

Exhaustive closeness ($O(V \cdot E)$) was the hang. We sampled BOTH measures (Eppstein–Wang).

Sampling ≠ wrong answer

Rank-correlation ≈0.99 with exact; ~9/10 top nodes identical. Rankings survive.

Parallel betweenness needs care

Naïve summing broke normalisation. We reused NetworkX rescaling — bit-identical to exact.

Girvan–Newman ≠ full graph

$O(E^2 \cdot V)$. Bounded to a structure-preserving ego subgraph; Louvain runs globally.

No leakage in link prediction

Features computed on the TRAIN graph only — test edges never see themselves.

SYNTHESIS

What the network told us

Scale-free topology

One giant component; mean degree 15, max 709 — a few hubs hold it together.

Influence is multi-faceted

Facebook bridges; the U.S. Army is locally dense. ρ (closeness,PageRank)=0.49.

Clear communities

60 Louvain groups at Q=0.808; Louvain beats GN 0.290 vs 0.112 on equal footing.

Predictable links

Topology alone gives AUC 0.956; degree-weighted shared neighbours dominate.

Thank you

A complete, reproducible graph-mining pipeline — tractable and correct at scale.

Code: github.com/alabyekkubo-ssuubi-brian/Graph-Mining

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